STRONG SPLITTING ON AN AFFINE BASE SHRINKS TO AN AFFINE INVARIANT OPEN

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; the author's verification sentence for (a), (b), (c) is an equality check written out in the source.

Allowed standard prerequisites (editorial).

- Scheme-version strong splitting. lemma-strong-splitting-scheme (carded L26661). Supplies the separated etale U' -> U and the point u' with P the strong splitting.
- Restriction preserves flatness and finite presentation. More on Groupoids, lemma restrict-preserves-type (cross-chapter, named). s', t' stay flat locally finitely presented after restriction.
- Quasi-affineness of quasi-finite separated morphisms into affines. More on Morphisms, lemma quasi-finite-separated-quasi-affine (cross-chapter, named). Zariski's Main Theorem form making W quasi-affine.
- Ample opens through finite sets and invariant affines. Properties, lemma ample-finite-set-in-affine; Groupoids, lemma find-invariant-affine (cross-chapter, named). Finds the affine P-invariant open U" containing u'.
- Finite flat is finite locally free. Morphisms, lemma finite-flat (cross-chapter, named). t(s^{-1}(u')) finite set of points.

Source reference. Original reference 03JE: focused-seeds/geometry/S9-smg-strong-splitting-affine-scheme/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03LN: focused-seeds/geometry/S9-smg-strong-splitting-affine-scheme/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04RX: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 2689–2689. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04MP: focused-seeds/geometry/S9-smg-strong-splitting-affine-scheme/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the

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Source reference. Original reference 02LR: focused-seeds/geometry/S9-smg-strong-splitting-affine-scheme/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03WS: focused-seeds/geometry/S9-smg-strong-splitting-affine-scheme/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02KB: focused-seeds/geometry/S9-smg-strong-splitting-affine-scheme/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01ZY: focused-seeds/geometry/S9-smg-strong-splitting-affine-scheme/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01K4: focused-seeds/geometry/S9-smg-strong-splitting-affine-scheme/ref-tags.txt, source lines 12–12. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04RL: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 2463–2463. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 04RL assume in addition that s, t are flat and locally of finite presentation and that U is affine. Then there exists an affine scheme U', an étale morphism $U' \to U$, and a point $u' \in U'$ lying over u with $\kappa(u) = \kappa(u')$ such that the restriction $R' = R|_{U'}$ of R to U' is strongly split over u'.*

Proof. Let $U' \to U$ and $u' \in U'$ be the separated étale morphism of schemes we found in Lemma 04RX. Let $P \subset R'$ be the strong splitting of R' over u'. By More on Groupoids, Lemma 04MP the morphisms $s', t' : R' \to U'$ are flat and locally of finite presentation. They are finite by assumption. Hence s', t' are finite locally free, see Morphisms, Lemma 02KB. In particular $t(s^{-1}(u'))$ is a finite set of points $\{u'_1, u'_2, \ldots, u'_n\}$ of U'. Choose a quasi-compact open $W \subset U'$ containing each u'_i. As U is affine the morphism $W \to U$ is quasi-compact (see Schemes, Lemma 01K4). The morphism $W \to U$ is also locally quasi-finite (see Morphisms, Lemma 03WS) and separated. Hence by More on Morphisms, Lemma 02LR (a version of Zariski's

Main Theorem) we conclude that W is quasi-affine. By Properties, Lemma 01ZY we see that $\{u'_1, \ldots, u'_n\}$ are contained in an affine open of U'. Thus we may apply Groupoids, Lemma 03JE to conclude that there exists an affine P-invariant open $U'' \subset U'$ which contains u'.

To finish the proof denote $R'' = R|_{U''}$ the restriction of R to U''. This is the same as the restriction of R' to U''. As $P \subset R'$ is an open and closed subscheme, so is $P|_{U''} \subset R''$. By construction the open subscheme $U'' \subset U'$ is P-invariant which means that $P|_{U''} = (s'|_P)^{-1}(U'') = (t'|_P)^{-1}(U'')$ (see discussion in Groupoids, Section 03LN) so the restrictions of s'' and t'' to $P|_{U''}$ are still finite. The sub groupoid scheme $P|_{U''}$ is still a strong splitting of R'' over u''; above we verified (a), (b) and (c) holds as $\{r' \in R' : t'(r') = u', s'(r') = u'\} = \{r'' \in R'' : t''(r'') = u', s''(r'') = u'\}$ trivially. The lemma is proved. $\square$